

DCS

copilot: advises, never writes to the process

macro- F_1 measured here

1 Perceive

window [21, 41)

alarm at 3σ

2 Score

the fold's 1D-CNN,
never retrained

3 Reason

21 documents, by
symptom similarity

rules, not an LLM

4 Decide

answer if $\text{conf} \geq 0.50$
else observe, escalate
or defer to the operator

Operator

approves
or rejects

plant

observe: the window advances 10 samples, 30 min of plant time, at most once